Week 4

Introduction

In the dynamic landscape of power electronics, the efficient and stable delivery of power is not merely an advantage but a fundamental requirement. As electric vehicle charging infrastructure expands and grid demands intensify, Power Factor Correction (PFC) converters have become indispensable. These devices are crucial for ensuring that electrical loads draw current efficiently from the AC mains, minimizing reactive power, and reducing harmonic distortion. This week, we delve into the sophisticated world of PFC converters, exploring the intricate details of their closed-loop control, the evolution of their topologies from conventional to advanced bridgeless designs, and alternative operating modes that offer unique trade-offs in performance and complexity.

Our journey begins with a deep dive into the control mechanisms that govern the behavior of single-phase Boost PFC converters, laying the groundwork for understanding how these systems regulate output voltage and achieve unity power factor. We then transition to the innovative realm of bridgeless PFC converters, examining how these designs overcome the inherent limitations of traditional topologies by reducing component count and conduction losses. The pinnacle of this evolution, the Totem Pole PFC converter, is explored in detail, highlighting its operational principles, efficiency advantages, and critical challenges like zero-crossing distortion. Finally, we investigate Discontinuous Conduction Mode (DCM) operation as a means to simplify control strategies, offering a different perspective on balancing performance with implementation complexity. Understanding these diverse approaches is vital for designing robust, efficient, and compliant power conversion systems for the future.

Core Concepts

Term	Definition	Significance
`Boost PFC Converter`	A power converter topology used to improve the power factor of an AC-DC power supply by shaping the input current.	Essential for meeting power quality standards and reducing harmonic distortion in the grid.
`Inner Current Loop`	The faster control loop responsible for regulating the	Ensures the input current is sinusoidal and in phase

Term	Definition	Significance
	inductor current to follow a desired reference waveform.	with the input voltage, achieving power factor correction.
`Totem Pole PFC`	A bridgeless power factor correction converter topology that eliminates the input diode bridge rectifier, using active switches.	Significantly reduces conduction losses and improves efficiency compared to traditional PFC circuits.
`DCM (Discontinuous Conduction Mode)`	An operating mode where the inductor current in a converter falls to zero during each switching cycle.	Simplifies control by eliminating the need for complex current loop feedback and multiplier blocks.
`Unity Power Factor`	A condition where the current drawn from the AC source is perfectly in phase with the voltage, resulting in minimal reactive power.	Essential for efficient grid utilization, reducing energy waste, and complying with power quality standards.
`Conduction Losses`	Power dissipated as heat in semiconductor devices when current flows through them due to on-state resistance or forward voltage drop.	A critical metric for converter efficiency; reducing these losses is a primary goal of advanced PFC topologies.
`Zero Crossing Distortion (ZCD)`	Undesirable spikes or irregularities in the input current waveform near the zero crossings of the input voltage.	A significant disadvantage of Totem Pole PFC, impacting Total Harmonic Distortion (THD) and power factor.
`Modulator Block`	The component that translates the control signal (duty ratio) into a pulse-width modulated (PWM) signal for switching devices.	Converts the continuous control command into discrete switching actions, enabling power conversion.

Term	Definition	Significance
`RMS Current`	The root mean square value of the current flowing through a component.	Important for selecting appropriate power devices and assessing conduction losses, especially in DCM.

Detailed Analysis

1. Closed-Loop Control of Conventional Boost PFC Converters

The **Closed Loop Control of Single-phase Boost PFC Converter-IV** lesson detailed the fundamental control architecture for single-phase Boost PFC converters, which primarily aims to regulate the output DC voltage and achieve unity power factor. This is accomplished through a nested two-loop control structure:
- **Outer Voltage Loop**: This slower loop regulates the DC output voltage to a set reference. It generates the current reference for the inner loop. The voltage controller (Cv(s)) requires a very high DC gain for precise regulation and a relatively low bandwidth (20-40 Hz) suitable for the slow-changing DC output.
- **Inner Current Loop**: This faster loop tracks the instantaneous inductor current reference, ensuring the input current is sinusoidal and in phase with the input voltage. The current controller (Cc(s)) must provide high gain at twice the line frequency (e.g., 100 Hz) to accurately track the varying reference and minimize error. Its bandwidth must be significantly less than the switching frequency (e.g., fsw/10) to attenuate high-frequency switching ripple.
- **Voltage Transfer Function (Gv(s))**: Derived using power balance and small-signal analysis, Gv(s) relates the output voltage to the inductor current. It is given by `Gv(s) = (Vs,peak / (2 * Vo)) * (RL / (RLCs + 1))`, highlighting the influence of input voltage, output voltage, load, and output capacitor.
- **Modulator Block**: This crucial component translates the control signal (duty ratio) into PWM signals for the converter switches by comparing the scaled duty ratio with a high-frequency carrier signal.
- **Stability Criteria**: Both loops require sufficient phase and gain margins (e.g., 35-60 degrees phase margin) to ensure stable and robust operation against disturbances.

2. Evolution to Bridgeless PFC Topologies

The **Bridgeless PFC Converter** and **Totem pole PFC Converter** lessons explored advancements aimed at overcoming the limitations of conventional Boost PFC

converters, particularly their high component count and significant conduction losses due to multiple semiconductor devices conducting simultaneously.
- **Limitations of Conventional Boost PFC**: Traditional designs use a full-bridge diode rectifier, leading to three devices typically conducting at any instant, resulting in higher conduction losses and increased bill of materials.
- **Standard Bridgeless PFC**: This topology eliminates the front-end diode bridge, integrating rectification into the boost stage. It uses two diodes and two active switches. While it reduces the number of conducting devices to two, its main disadvantage is that the diodes are forced to switch at the high switching frequency (fsw), leading to significant reverse recovery losses and requiring expensive fast recovery diodes.
- **Totem Pole PFC (Diode-Assisted)**: This advanced bridgeless topology intelligently arranges switches and diodes to mitigate high-frequency diode switching losses. It features an active leg (S1, S2) switching at high frequency and a slow-switching leg (D1, D2) operating at line frequency.
- In the positive half-cycle, D2 conducts continuously, while S1 and S2 switch at high frequency.
- In the negative half-cycle, D1 conducts continuously, while S1 and S2 switch at high frequency.
- The duty ratio `D = 1 - |Vs,pk sin(ωt)| / V0` is applied to S2 in the positive half-cycle and S1 in the negative half-cycle.
- This design significantly reduces high-frequency diode losses as D1 and D2 only switch at line frequency.
- **Enhanced Totem Pole PFC (All-MOSFET)**: To further improve efficiency, the passive diodes (D1, D2) in the slow-switching leg are replaced by active switches (S3, S4), typically MOSFETs. These MOSFETs are actively turned on to utilize their low-resistance channels, drastically reducing conduction losses compared to diodes.
- The fast-switching leg (S1, S2) operates at high frequency (e.g., 10-40 kHz).
- The slow-switching leg (S3, S4) operates at line frequency (e.g., 50/60 Hz), contributing minimal switching losses.
- This configuration ensures only two devices are in the conduction path at any time, leading to very high efficiency (often >98%).
- Component sizing for the inductor and capacitor follows principles similar to conventional boost PFC. All switches must be rated for at least `1.4 * V0` for voltage and their RMS current ratings must be carefully calculated based on their specific conduction periods and duty cycles.

3. Control and Challenges of Totem Pole PFC

The **Totem pole PFC Converter-II** lesson elaborated on the closed-loop control strategy and a significant challenge of the Totem Pole PFC.

- **Closed-Loop Control Strategy**: Similar to conventional PFC, it employs an average current mode control. An outer voltage loop regulates V_o, generating a current magnitude reference. This magnitude is multiplied by a unit sine wave (sin ωt) in phase with the input grid voltage to create the reference input current (i_s,ref). An inner current loop then forces the actual inductor current (i_L) to track i_s,ref.
- **Modulator Block Operation**: The modulator generates PWM signals for all four switches (S1, S2, S3, S4) based on the calculated duty ratio `d(t)` and the polarity of the input voltage `vs(t)`.
- For `vs > 0`: `dS1(t) = 1 - d(t)`, `dS2(t) = d(t)`, `dS3(t) = 0` (off), `dS4(t) = 1` (on).
- For `vs < 0`: `dS1(t) = d(t)`, `dS2(t) = 1 - d(t)`, `dS3(t) = 1` (on), `dS4(t) = 0` (off).
- **Zero Crossing Distortion (ZCD)**: A major disadvantage of the Totem Pole PFC is the occurrence of ZCD in the input current, particularly near the zero crossings of the input voltage.
- **Causes**: This distortion arises from non-ideal switch behavior, delays in turn-on/off (especially S4), and abrupt changes in duty ratios at the AC line voltage crossover points.
- **Impact**: These issues can lead to large positive or negative inductor current spikes, causing the input current to deviate significantly from a pure sine wave. This results in higher Total Harmonic Distortion (THD) and a reduced power factor, especially problematic at light load conditions.

4. Discontinuous Conduction Mode (DCM) Operation

The **DCM operation of Boost PFC Converter** lesson introduced an alternative operating mode that simplifies control at the cost of some performance metrics.
- **Limitations of Continuous Conduction Mode (CCM)**: CCM PFC requires complex control, including multiplier blocks to shape the current reference and accurate generation of unit sine waveforms synchronized with the grid. This leads to high computational burden and increased cost.
- **DCM Principle**: In DCM, the inductor current is intentionally allowed to fall to zero within every switching cycle. After reaching zero, it remains at zero until the next switching cycle begins.
- **Advantages of DCM**:
- **Automatic Current Following**: The peak inductor current inherently follows the modulus of the input AC voltage, leading to a "voltage follower mode" where the input current automatically approximates a sinusoidal shape.
- **Simplified Control**: This eliminates the need for complex multiplier blocks and precise unit sine waveform generation, significantly reducing computational

requirements and simplifying the control circuit. Often, only a single voltage control loop is sufficient.
- **Reduced Switching Losses**: Switches turn on with zero current (ZCS), and diodes turn off with zero current, minimizing reverse recovery losses.
- **Disadvantages of DCM**:
- **Higher THD**: The input current is only "close to sinusoidal," resulting in higher Total Harmonic Distortion compared to CCM.
- **Increased Ripple and RMS Currents**: DCM operation leads to larger peak-to-peak inductor current ripple and higher RMS currents through the inductor and switches, increasing conduction losses and requiring higher-rated components.
- **Higher S1 Turn-Off Losses**: While S1 turns on with ZCS, it turns off with the peak inductor current, potentially leading to higher turn-off switching losses.

Key Takeaways

- [ ] Closed-loop control of Boost PFC converters uses nested inner current and outer voltage loops for regulation and power factor correction.
- [ ] The inner current loop requires high gain at 2fs and bandwidth less than fsw/10 for accurate tracking and ripple attenuation.
- [ ] Conventional Boost PFC suffers from high conduction losses due to multiple conducting devices and a diode bridge rectifier.
- [ ] Bridgeless PFC topologies reduce conduction losses by minimizing the number of conducting devices in the current path.
- [ ] The Totem Pole PFC converter achieves high efficiency by having active switches (S1, S2) operate at high frequency and diodes (or S3, S4) operate at line frequency.
- [ ] Replacing diodes with MOSFETs (S3, S4) in the Totem Pole PFC further reduces conduction losses, leading to very high overall efficiency.
- [ ] Zero Crossing Distortion (ZCD) is a significant challenge in Totem Pole PFC, caused by switch delays and abrupt duty ratio changes, leading to increased THD.
- [ ] DCM operation simplifies PFC control by allowing the inductor current to drop to zero, eliminating multiplier blocks and unit sine wave generation.
- [ ] DCM offers reduced switching losses (ZCS, zero reverse recovery) but results in higher THD and RMS currents compared to CCM.
- [ ] Component sizing (inductor, capacitor, switch voltage/current ratings) is critical for robust and efficient operation across all PFC topologies.

Conclusion

This week's exploration into the control and topologies of single-phase Boost PFC converters underscores the continuous innovation driving power electronics. From the foundational two-loop control of conventional designs to the highly efficient bridgeless and Totem Pole architectures, the overarching goal remains consistent: to deliver stable DC power while drawing current efficiently from the AC grid. The evolution towards bridgeless topologies, particularly the Totem Pole PFC, represents a significant leap in minimizing conduction and switching losses, crucial for high-power applications like EV charging where every percentage point of efficiency matters.

However, these advancements are not without their complexities, as evidenced by the challenge of zero-crossing distortion in Totem Pole converters. This highlights the intricate trade-offs designers face between maximizing efficiency and maintaining pristine power quality. The introduction of Discontinuous Conduction Mode (DCM) offers a compelling alternative, prioritizing control simplicity and cost reduction, albeit with compromises in harmonic performance. Ultimately, the choice of PFC topology and operating mode is dictated by the specific application requirements, balancing the need for high power factor, low harmonic distortion, and robust output voltage regulation against considerations of cost, control complexity, and component stress. As the demand for cleaner and more efficient power conversion grows, a deep understanding of these diverse strategies will be paramount for shaping the next generation of power systems.